

DESIGN AND ANALYSIS OF ALGORITHMS

DAY 4 – DYNAMIC PROGRAMMING – BASIC PROBLEMS

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Course Name: Design and Analysis of Algorithm

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Experiment 1: Dice Throw – Reward Score

Objective: Determine the number of ways to obtain an exact target score using a given number of dice.

Problem Statement: Determine the number of ways to obtain an exact target score using a given number of dice.

Sample Input: 3 dice, 6 faces per die, target score 8

Sample Output: 21

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 3
Enter faces per die: 6
Enter target sum: 8
Number of ways = 21
```

Result: The program was implemented and the expected result was obtained.

Experiment 2: Dice Throw – Board Game Tournament

Objective: Calculate the number of outcomes when 2 dice must produce a target score of 7.

Problem Statement: Calculate the number of outcomes when 2 dice must produce a target score of 7.

Sample Input: 2 dice, 6 faces per die, target score 7

Sample Output: 6

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 2
Enter faces per die: 6
Enter target sum: 7
Number of ways = 6
```

Result: The program was implemented and the expected result was obtained.

Experiment 3: Dice Throw – Warehouse Robot Navigation

Objective: Determine the number of movement sequences when 4 dice with 4 faces have a target sum of 10.

Problem Statement: Determine the number of movement sequences when 4 dice with 4 faces have a target sum of 10.

Sample Input: 4 dice, 4 faces per die, target distance 10

Sample Output: 44

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 4
Enter faces per die: 4
Enter target sum: 10
Number of ways = 44
```

Result: The program was implemented and the expected result was obtained.

Experiment 4: Dice Throw – Casino Reward

Objective: Calculate the winning combinations when 3 six-faced dice produce a target sum of 12.

Problem Statement: Calculate the winning combinations when 3 six-faced dice produce a target sum of 12.

Sample Input: 3 dice, 6 faces per die, target sum 12

Sample Output: 25

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
            dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 3
Enter faces per die: 6
Enter target sum: 12
Number of ways = 25
```

Result: The program was implemented and the expected result was obtained.

Experiment 5: Dice Throw – Quiz Scoring

Objective: Find the number of valid scoring combinations when 5 six-faced dice produce a target score of 15.

Problem Statement: Find the number of valid scoring combinations when 5 six-faced dice produce a target score of 15.

Sample Input: 5 dice, 6 faces per die, target score 15

Sample Output: 651

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 5
Enter faces per die: 6
Enter target sum: 15
Number of ways = 651
```

Result: The program was implemented and the expected result was obtained.

Experiment 6: Dice Throw – Sports Prediction

Objective: Determine the number of ways to obtain a predicted score of 9 using 3 five-faced dice.

Problem Statement: Determine the number of ways to obtain a predicted score of 9 using 3 five-faced dice.

Sample Input: 3 dice, 5 faces per die, target score 9

Sample Output: 19

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 3
Enter faces per die: 5
Enter target sum: 9
Number of ways = 19
```

Result: The program was implemented and the expected result was obtained.

Experiment 7: Dice Throw – Lottery Simulation

Objective: Find the number of combinations from 2 ten-faced dice that produce a sum of 11.

Problem Statement: Find the number of combinations from 2 ten-faced dice that produce a sum of 11.

Sample Input: 2 dice, 10 faces per die, target sum 11

Sample Output: 10

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 2
Enter faces per die: 10
Enter target sum: 11
Number of ways = 10
```

Result: The program was implemented and the expected result was obtained.

Experiment 8: Dice Throw – General N Dice

Objective: Find the number of ways to obtain a given sum using N dice, each having M faces.

Problem Statement: Find the number of ways to obtain a given sum using N dice, each having M faces.

Sample Input: 2 dice, 6 faces per die, target sum 7

Sample Output: 6

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 2
Enter faces per die: 6
Enter target sum: 7
Number of ways = 6
```

Result: The program was implemented and the expected result was obtained.

Experiment 9: Dice Throw – Target Score

Objective: Determine the number of ways to obtain a target score using multiple dice.

Problem Statement: Determine the number of ways to obtain a target score using multiple dice.

Sample Input: 3 dice, 6 faces per die, target sum 8

Sample Output: 21

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 3
Enter faces per die: 6
Enter target sum: 8
Number of ways = 21
```

Result: The program was implemented and the expected result was obtained.

Experiment 10: Dice Throw – Specified Sum

Objective: Count possible outcomes that produce a specified sum.

Problem Statement: Count possible outcomes that produce a specified sum.

Sample Input: 4 dice, 6 faces per die, target sum 10

Sample Output: 80

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 4
Enter faces per die: 6
Enter target sum: 10
Number of ways = 80
```

Result: The program was implemented and the expected result was obtained.

Experiment 11: Dice Throw – Required Score

Objective: Find the number of dice combinations that yield a required score.

Problem Statement: Find the number of dice combinations that yield a required score.

Sample Input: 2 dice, 4 faces per die, target sum 5

Sample Output: 4

Time Complexity: $O(D \times T \times F)$ time and $O(T)$ auxiliary space.

Explanation: Dynamic programming stores the number of ways to reach each possible sum after every die, avoiding repeated enumeration of partial sums.

Python Program

```
def count_ways(dice, faces, target):
    dp = [0] * (target + 1)
    dp[0] = 1

    for _ in range(dice):
        next_dp = [0] * (target + 1)
        for total in range(target + 1):
            for face in range(1, faces + 1):
                if total + face <= target:
                    next_dp[total + face] += dp[total]
        dp = next_dp

    return dp[target]

dice = int(input("Enter number of dice: "))
faces = int(input("Enter faces per die: "))
target = int(input("Enter target sum: "))
print("Number of ways =", count_ways(dice, faces, target))
```

Execution / Output

```
Enter number of dice: 2
Enter faces per die: 4
Enter target sum: 5
Number of ways = 4
```

Result: The program was implemented and the expected result was obtained.

Experiment 12: Dice Throw – Binomial Section

Objective: Determine the number of ways to obtain a target sum using dice throws.

Problem Statement: Determine the number of ways to obtain a target sum using dice throws.

Sample Input: 3 dice, 4 faces per die, target sum 7

Sample Output: 12

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter number of dice: 3
Enter faces per die: 4
Enter target sum: 7
Number of ways = 12
```

Result: The program was implemented and the expected result was obtained.

Experiment 13: Binomial Coefficient $C(n,r)$

Objective: Calculate the Binomial Coefficient $C(n,r)$.

Problem Statement: Calculate the Binomial Coefficient $C(n,r)$.

Sample Input: $n = 5, r = 2$

Sample Output: $C(5,2) = 10$

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 5
Enter r: 2
C(5,2) = 10
```

Result: The program was implemented and the expected result was obtained.

Experiment 14: Selection of r Items

Objective: Find the number of ways to select r items from n items.

Problem Statement: Find the number of ways to select r items from n items.

Sample Input: $n = 8, r = 3$

Sample Output: $C(8,3) = 56$

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 8
Enter r: 3
C(8,3) = 56
```

Result: The program was implemented and the expected result was obtained.

Experiment 15: Binomial Coefficient using Dynamic Programming

Objective: Compute the Binomial Coefficient using Dynamic Programming.

Problem Statement: Compute the Binomial Coefficient using Dynamic Programming.

Sample Input: $n = 10, r = 4$

Sample Output: $C(10,4) = 210$

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 10
Enter r: 4
C(10,4) = 210
```

Result: The program was implemented and the expected result was obtained.

Experiment 16: Committee Formation

Objective: Determine the number of possible committees formed from a group.

Problem Statement: Determine the number of possible committees formed from a group.

Sample Input: Total members = 12, committee size = 5

Sample Output: 792

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 12
Enter r: 5
792
```

Result: The program was implemented and the expected result was obtained.

Experiment 17: Combination Value

Objective: Calculate the combination value for given n and r.

Problem Statement: Calculate the combination value for given n and r.

Sample Input: n = 15, r = 6

Sample Output: C(15,6) = 5005

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 15
Enter r: 6
C(15,6) = 5005
```

Result: The program was implemented and the expected result was obtained.

Experiment 18: AI Project Team Selection

Objective: Calculate the number of different teams formed by selecting developers.

Problem Statement: Calculate the number of different teams formed by selecting developers.

Sample Input: 10 developers, 3 required

Sample Output: 120

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 10
Enter r: 3
120
```

Result: The program was implemented and the expected result was obtained.

Experiment 19: Scholarship Committee

Objective: Determine the number of committees formed by selecting faculty members.

Problem Statement: Determine the number of committees formed by selecting faculty members.

Sample Input: 12 professors, committee size 4

Sample Output: 495

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 12
Enter r: 4
495
```

Result: The program was implemented and the expected result was obtained.

Experiment 20: Cricket Tournament Squad

Objective: Calculate the possible selections of players for a tournament squad.

Problem Statement: Calculate the possible selections of players for a tournament squad.

Sample Input: 15 players, 5 required

Sample Output: 3003

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 15
Enter r: 5
3003
```

Result: The program was implemented and the expected result was obtained.

Experiment 21: Product Testing Groups

Objective: Determine the number of testing groups formed from volunteers.

Problem Statement: Determine the number of testing groups formed from volunteers.

Sample Input: 20 volunteers, group size 6

Sample Output: 38760

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 20
Enter r: 6
38760
```

Result: The program was implemented and the expected result was obtained.

Experiment 22: Server Combinations

Objective: Determine the possible combinations of servers allocated to a customer.

Problem Statement: Determine the possible combinations of servers allocated to a customer.

Sample Input: 8 servers, 3 selected

Sample Output: 56

Time Complexity: $O(n \times r)$ time and $O(r)$ auxiliary space using one-dimensional dynamic programming.

Explanation: Pascal's identity is used to build the combination values. Updating the table from right to left preserves values from the previous stage.

Python Program

```
def binomial(n, r):
    if r < 0 or r > n:
        return 0

    r = min(r, n - r)
    dp = [0] * (r + 1)
    dp[0] = 1

    for i in range(1, n + 1):
        for j in range(min(i, r), 0, -1):
            dp[j] += dp[j - 1]

    return dp[r]

n = int(input("Enter n: "))
r = int(input("Enter r: "))
print(f"C({n},{r}) =", binomial(n, r))
```

Execution / Output

```
Enter n: 8
Enter r: 3
56
```

Result: The program was implemented and the expected result was obtained.

Experiment 23: Assembly Line Scheduling

Objective: Find the minimum time required to manufacture a product using two assembly lines.

Problem Statement: Find the minimum time required to manufacture a product using two assembly lines.

Sample Input: [10,12]; [18,7]; L1 [4,5,3,2]; L2 [2,10,1,4]; T1 [7,4,5]; T2 [9,2,8]

Sample Output: 35

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 10 12
Enter exit times: 18 7
Enter Line 1 times: 4 5 3 2
Enter Line 2 times: 2 10 1 4
Enter Transfer 1 times: 7 4 5
Enter Transfer 2 times: 9 2 8
Minimum Production Time = 35
```

Result: The program was implemented and the expected result was obtained.

Experiment 24: Optimal Assembly Line Path

Objective: Determine the optimal assembly line path with minimum processing time.

Problem Statement: Determine the optimal assembly line path with minimum processing time.

Sample Input: [8,10]; [12,15]; L1 [5,6,4]; L2 [7,3,5]; T1 [2,1]; T2 [3,2]

Sample Output: 33

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                    time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                    time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 8 10
Enter exit times: 12 15
Enter Line 1 times: 5 6 4
Enter Line 2 times: 7 3 5
Enter Transfer 1 times: 2 1
Enter Transfer 2 times: 3 2
Minimum Production Time = 33
```

Result: The program was implemented and the expected result was obtained.

Experiment 25: Fastest Route Through Stations

Objective: Compute the fastest route through assembly stations.

Problem Statement: Compute the fastest route through assembly stations.

Sample Input: [5,6]; [4,5]; L1 [3,4,5]; L2 [4,3,6]; T1 [2,1]; T2 [1,2]

Sample Output: 21

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                    time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                    time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 5 6
Enter exit times: 4 5
Enter Line 1 times: 3 4 5
Enter Line 2 times: 4 3 6
Enter Transfer 1 times: 2 1
Enter Transfer 2 times: 1 2
Minimum Production Time = 21
```

Result: The program was implemented and the expected result was obtained.

Experiment 26: Minimum Manufacturing Schedule

Objective: Identify the assembly line schedule that minimizes total manufacturing time.

Problem Statement: Identify the assembly line schedule that minimizes total manufacturing time.

Sample Input: [7,9]; [8,6]; L1 [4,3,5,2]; L2 [5,2,4,3]; T1 [1,2,1]; T2 [2,1,2]

Sample Output: 29

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 7 9
Enter exit times: 8 6
Enter Line 1 times: 4 3 5 2
Enter Line 2 times: 5 2 4 3
Enter Transfer 1 times: 1 2 1
Enter Transfer 2 times: 2 1 2
Minimum Production Time = 29
```

Result: The program was implemented and the expected result was obtained.

Experiment 27: Minimum Cost Assembly Path

Objective: Determine the minimum cost path through an assembly system.

Problem Statement: Determine the minimum cost path through an assembly system.

Sample Input: [6,7]; [5,4]; L1 [2,4,3]; L2 [3,2,5]; T1 [1,2]; T2 [2,1]

Sample Output: 18

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 6 7
Enter exit times: 5 4
Enter Line 1 times: 2 4 3
Enter Line 2 times: 3 2 5
Enter Transfer 1 times: 1 2
Enter Transfer 2 times: 2 1
Minimum Production Time = 18
```

Result: The program was implemented and the expected result was obtained.

Experiment 28: Automobile Manufacturing

Objective: Determine the minimum time required to manufacture a car using two assembly lines.

Problem Statement: Determine the minimum time required to manufacture a car using two assembly lines.

Sample Input: [10,12]; [18,7]; L1 [4,5,3,2]; L2 [2,10,1,4]; T1 [7,4,5]; T2 [9,2,8]

Sample Output: 35

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 10 12
Enter exit times: 18 7
Enter Line 1 times: 4 5 3 2
Enter Line 2 times: 2 10 1 4
Enter Transfer 1 times: 7 4 5
Enter Transfer 2 times: 9 2 8
Minimum Production Time = 35
```

Result: The program was implemented and the expected result was obtained.

Experiment 29: Smartphone Production

Objective: Determine the optimal production route when components may switch lines.

Problem Statement: Determine the optimal production route when components may switch lines.

Sample Input: [8,10]; [12,15]; L1 [5,6,4]; L2 [7,3,5]; T1 [2,1]; T2 [3,2]

Sample Output: 33

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 8 10
Enter exit times: 12 15
Enter Line 1 times: 5 6 4
Enter Line 2 times: 7 3 5
Enter Transfer 1 times: 2 1
Enter Transfer 2 times: 3 2
Minimum Production Time = 33
```

Result: The program was implemented and the expected result was obtained.

Experiment 30: Food Packaging Lines

Objective: Find the minimum completion time through labeling, sealing, and packing stations.

Problem Statement: Find the minimum completion time through labeling, sealing, and packing stations.

Sample Input: [5,6]; [4,5]; L1 [3,4,5]; L2 [4,3,6]; T1 [2,1]; T2 [1,2]

Sample Output: 21

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 5 6
Enter exit times: 4 5
Enter Line 1 times: 3 4 5
Enter Line 2 times: 4 3 6
Enter Transfer 1 times: 2 1
Enter Transfer 2 times: 1 2
Minimum Production Time = 21
```

Result: The program was implemented and the expected result was obtained.

Experiment 31: Medicine Bottle Production

Objective: Determine the optimal production schedule through two assembly lines.

Problem Statement: Determine the optimal production schedule through two assembly lines.

Sample Input: [7,9]; [8,6]; L1 [4,3,5,2]; L2 [5,2,4,3]; T1 [1,2,1]; T2 [2,1,2]

Sample Output: 29

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 7 9
Enter exit times: 8 6
Enter Line 1 times: 4 3 5 2
Enter Line 2 times: 5 2 4 3
Enter Transfer 1 times: 1 2 1
Enter Transfer 2 times: 2 1 2
Minimum Production Time = 29
```

Result: The program was implemented and the expected result was obtained.

Experiment 32: Circuit Board Manufacturing

Objective: Find the minimum manufacturing time when products can switch between two assembly lines.

Problem Statement: Find the minimum manufacturing time when products can switch between two assembly lines.

Sample Input: [6,7]; [5,4]; L1 [2,4,3]; L2 [3,2,5]; T1 [1,2]; T2 [2,1]

Sample Output: 18

Time Complexity: $O(n)$ time and $O(1)$ auxiliary space.

Explanation: At each station, the minimum time for each line is calculated by considering staying on that line or transferring from the other line.

Python Program

```
def assembly_line(entry, exit_time, line1, line2, transfer1, transfer2):
    time1 = entry[0] + line1[0]
    time2 = entry[1] + line2[0]

    for i in range(1, len(line1)):
        new1 = min(time1 + line1[i],
                   time2 + transfer2[i - 1] + line1[i])
        new2 = min(time2 + line2[i],
                   time1 + transfer1[i - 1] + line2[i])
        time1, time2 = new1, new2

    return min(time1 + exit_time[0], time2 + exit_time[1])

entry = list(map(int, input("Enter entry times: ").split()))
exit_time = list(map(int, input("Enter exit times: ").split()))
line1 = list(map(int, input("Enter Line 1 times: ").split()))
line2 = list(map(int, input("Enter Line 2 times: ").split()))
transfer1 = list(map(int, input("Enter Transfer 1 times: ").split()))
transfer2 = list(map(int, input("Enter Transfer 2 times: ").split()))

print("Minimum Production Time =", assembly_line(
    entry, exit_time, line1, line2, transfer1, transfer2
))
```

Execution / Output

```
Enter entry times: 6 7
Enter exit times: 5 4
Enter Line 1 times: 2 4 3
Enter Line 2 times: 3 2 5
Enter Transfer 1 times: 1 2
Enter Transfer 2 times: 2 1
Minimum Production Time = 18
```

Result: The program was implemented and the expected result was obtained.